

1.3 Principles of Design for Environment

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What this lecture covers



Design for Environment
(DFE) definition



Key steps to establish a
successful DFE program



The seven principles
of DFE

Design for Environment (DFE) Definition

The systematic consideration of design performance with respect to environmental, health, safety, and sustainability objectives over the full product and process life cycle.



What Prevents DFE Implementation?

- Limited resources for new projects
- Organisational and cultural inertia favours *business-as-usual*
- Environmental issues poorly understood by managers and many engineers
- Existing accounting systems do not reflect environmental value, difficult to quantify
- Engineers fear compromising product quality or production efficiency and introducing complexity

Key Steps to Establish Successful DFE



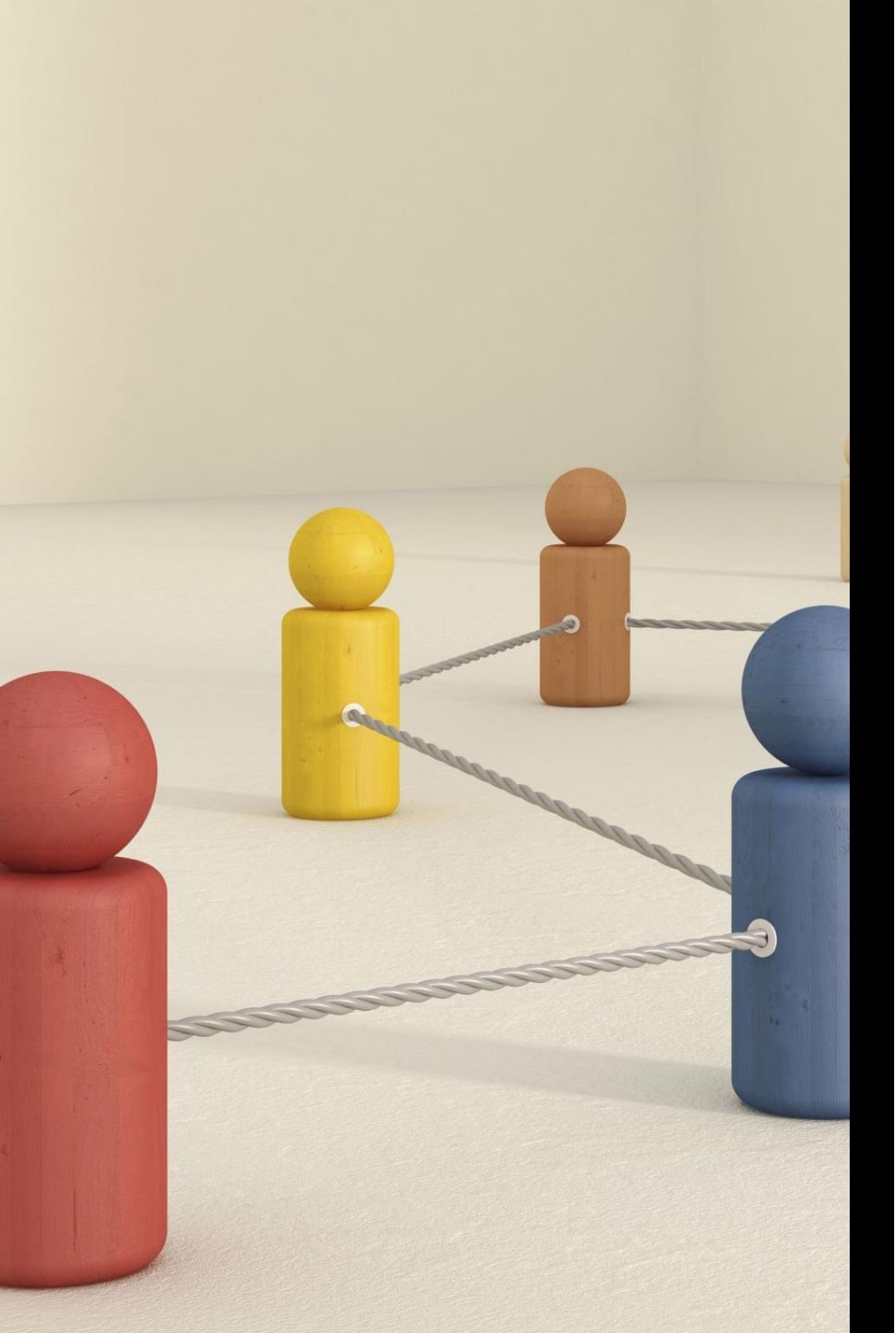
PROGRAM
DEFINITION



DFE
IMPLEMENTATION



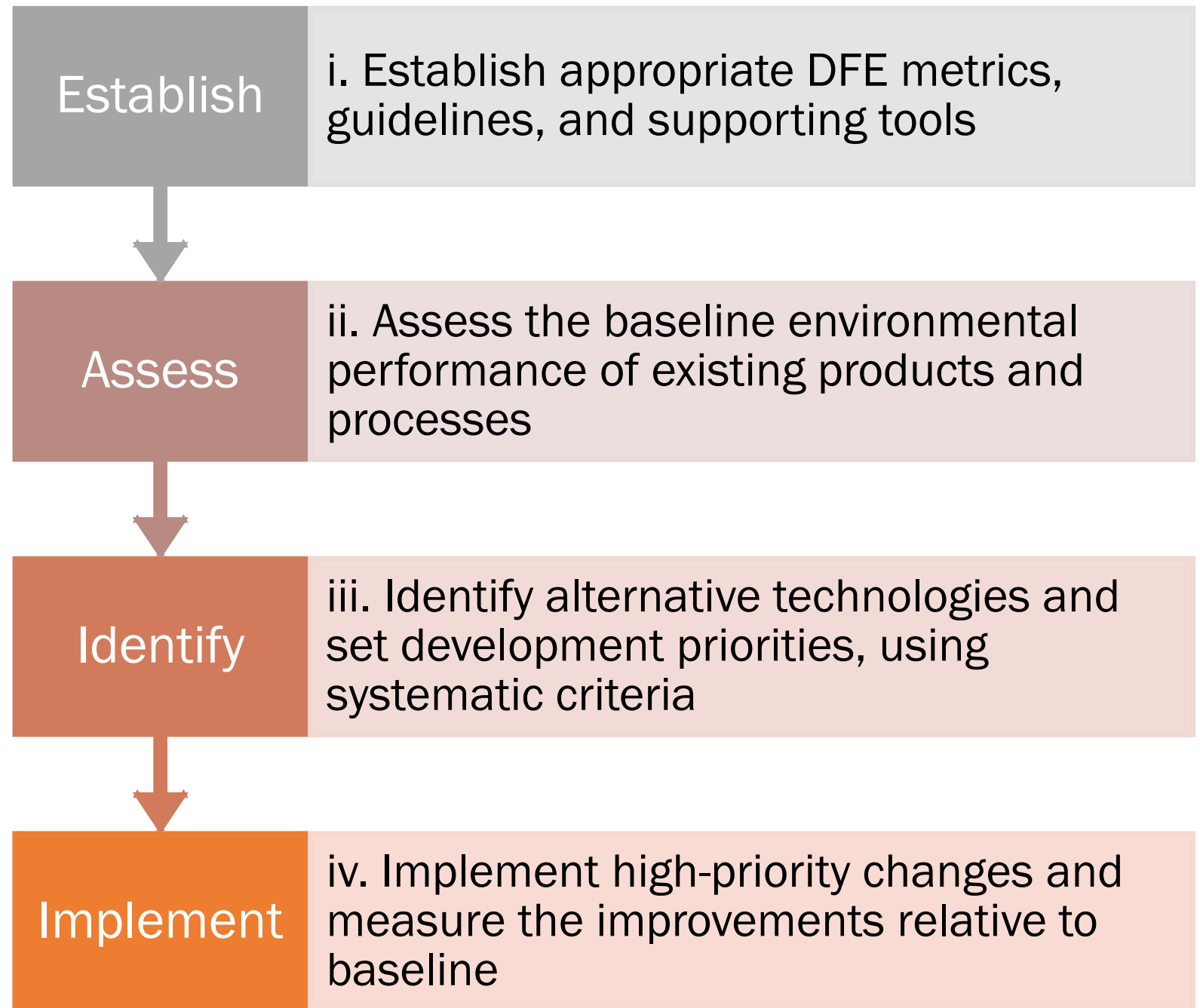
ONGOING
DFE



1. Program Definition and Targets

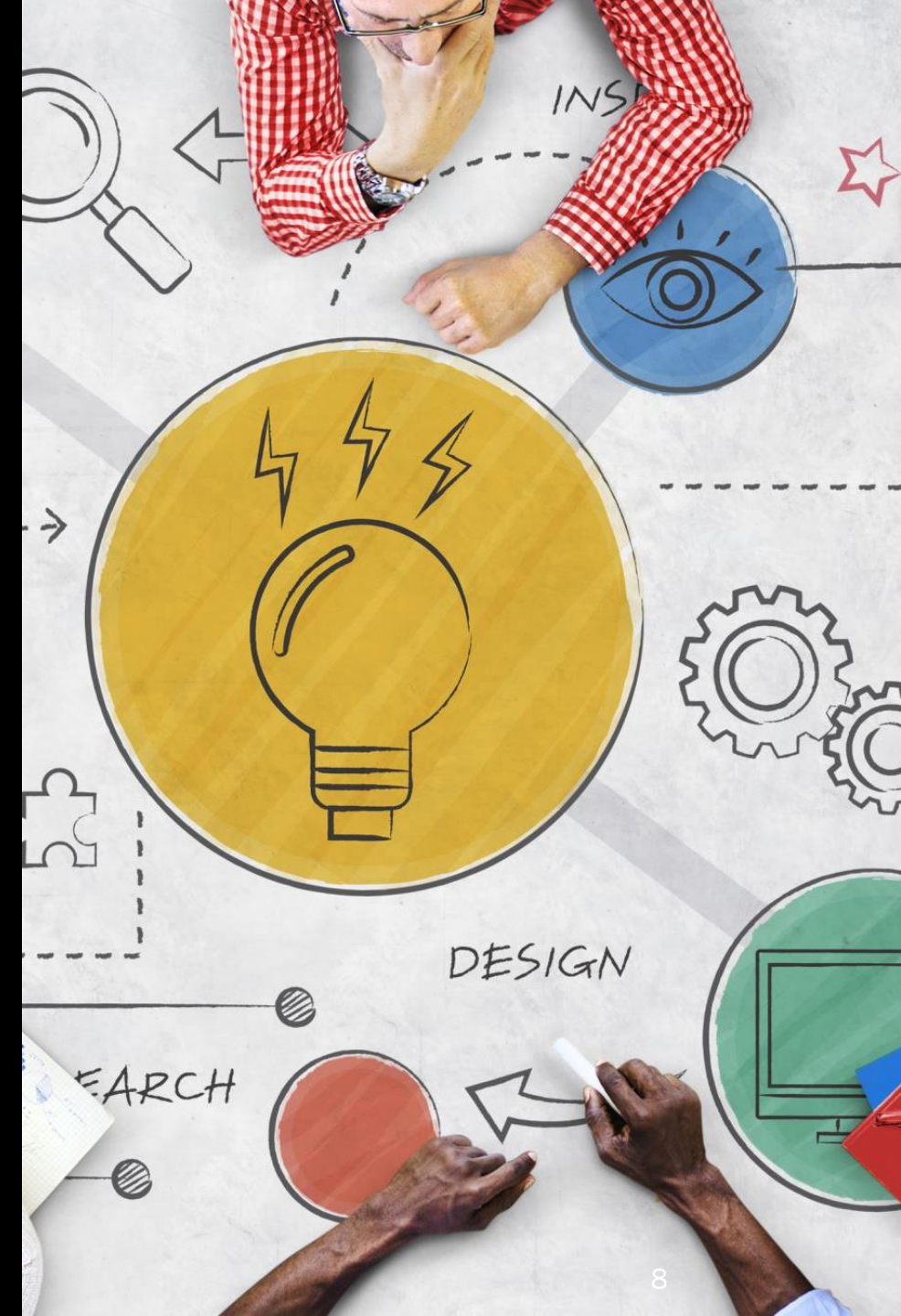
- i. Establish top management commitment (e.g., targets) to legitimise the program
- ii. Invite inputs from external stakeholders, e.g., regulators, customers, Māori, and communities
- iii. Integrate the program within the existing organisational structure and cross-functional teams
- iv. Ensure adequate staff training, incentives, and empowerment

2. DFE Plan & Implementation

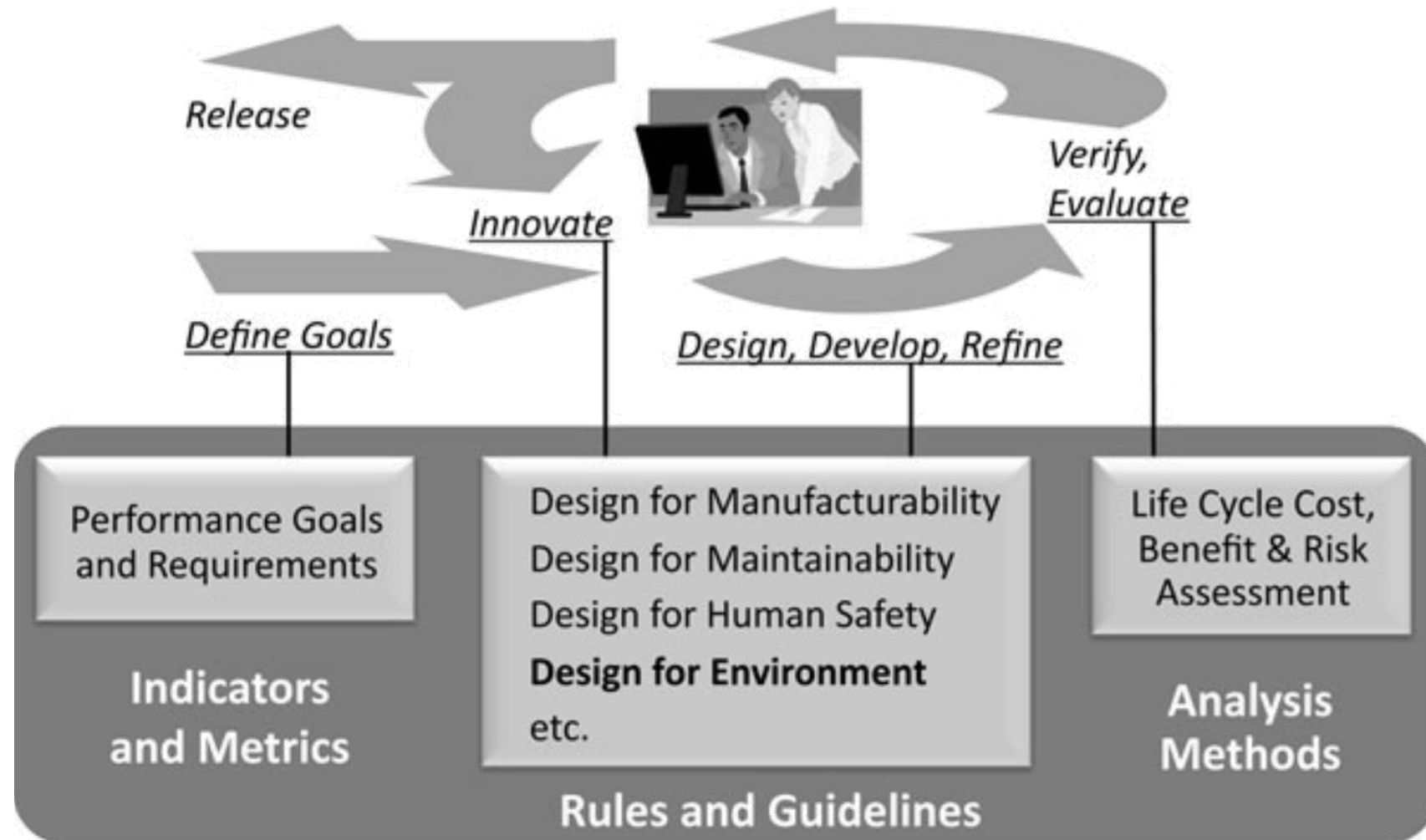


3. Ongoing DFE

- i. Integrate new technologies and lessons learned into future outputs
- ii. Institutionalise the metrics, guidelines and other tools
- iii. Communicate the beneficial results of DFE actions to stakeholders
- iv. Recognise and reward both team and individual accomplishments



Incorporating DFE into Design



Examples of Business DFE Strategies

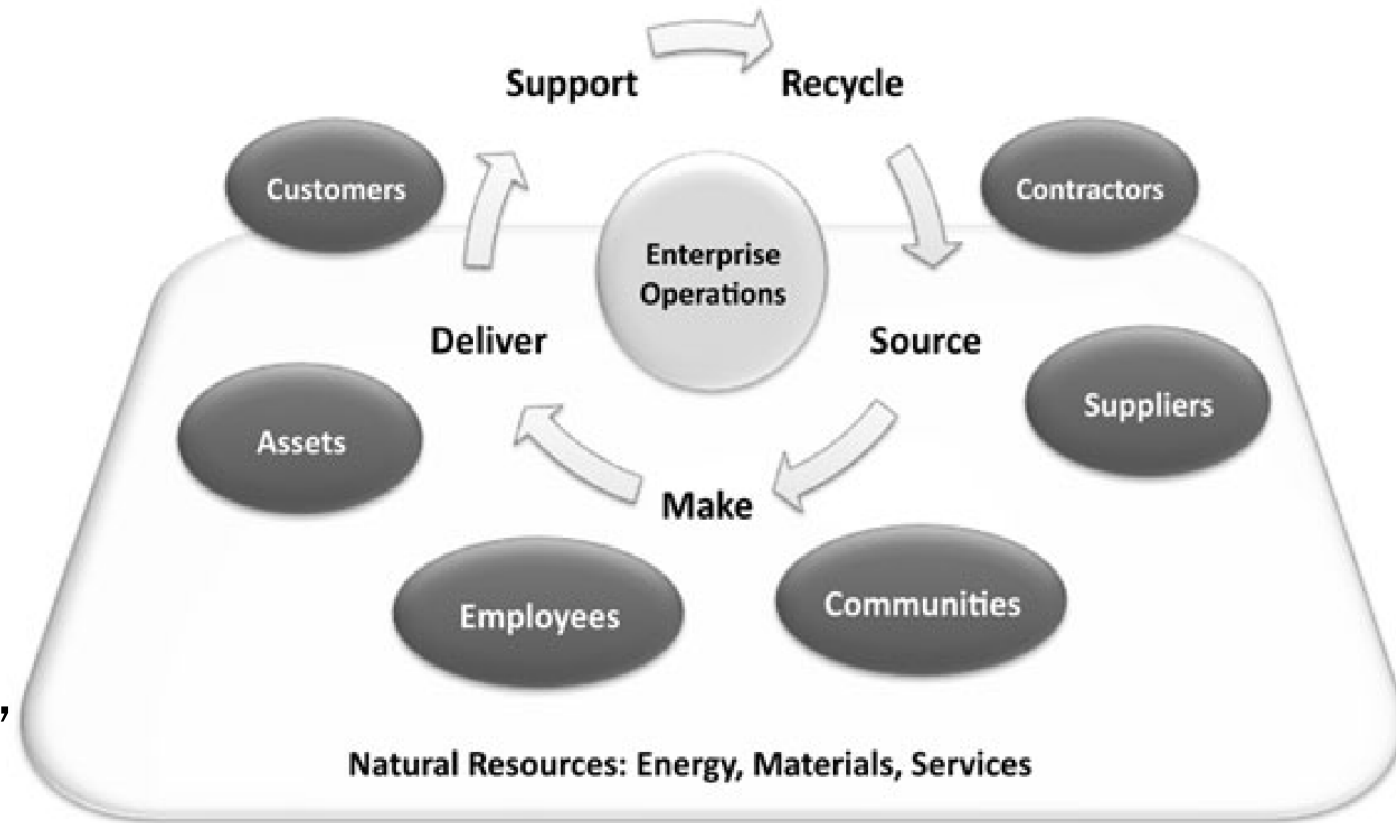
- **Intel** has worked closely with its customers to develop eco-efficient, reusable packaging for transporting microchips through the different stages of manufacturing and assembly.
- **Coca-Cola** has teamed with other food and beverage companies to encourage more rapid development and adoption of sustainable refrigeration technologies.
- **General Motors** has worked with the U.S. Environmental Protection Agency's Green Suppliers Network to encourage environmentally beneficial practices on the part of automotive suppliers.

The Seven Principles of DFE

1. Embed **life-cycle thinking** into the engineering process
2. Evaluate the **material and energy efficiency** and effectiveness of the overall system
3. Select appropriate **metrics** to represent product life-cycle performance
4. Maintain and apply a portfolio of systematic **design strategies**
5. Use **analysis methods** to evaluate design performance and understand trade-offs
6. Provide **software** capabilities to facilitate the application of DFE practices
7. Seek **inspiration from nature** for the design of products and systems

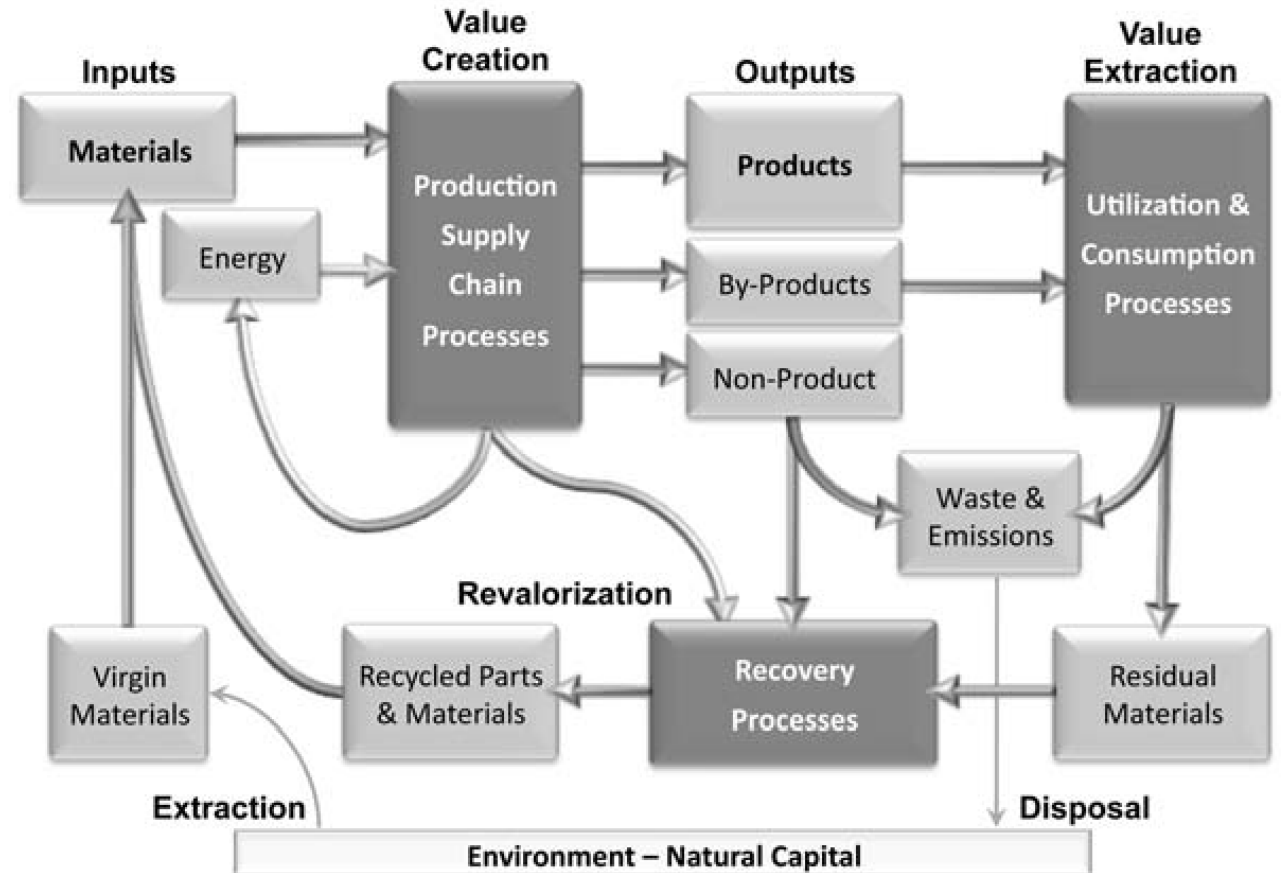
1. Life-Cycle Thinking

- ❑ The fundamental to life-cycle assessment and the related analysis methods
- ❑ Each life-cycle stage consumes natural resources, generates environmental impacts, and creates costs or benefits for stakeholders
- ❑ It considers a variety of impacts, e.g., economic, environmental, and social



2. Material and Energy Efficiency

- Derive economic value creation from environmental resource protection.
- Need a resource productivity to correlate with profitability
- Require a clear “system boundary” (physical processes, time)



3. Life-cycle Performance Metrics

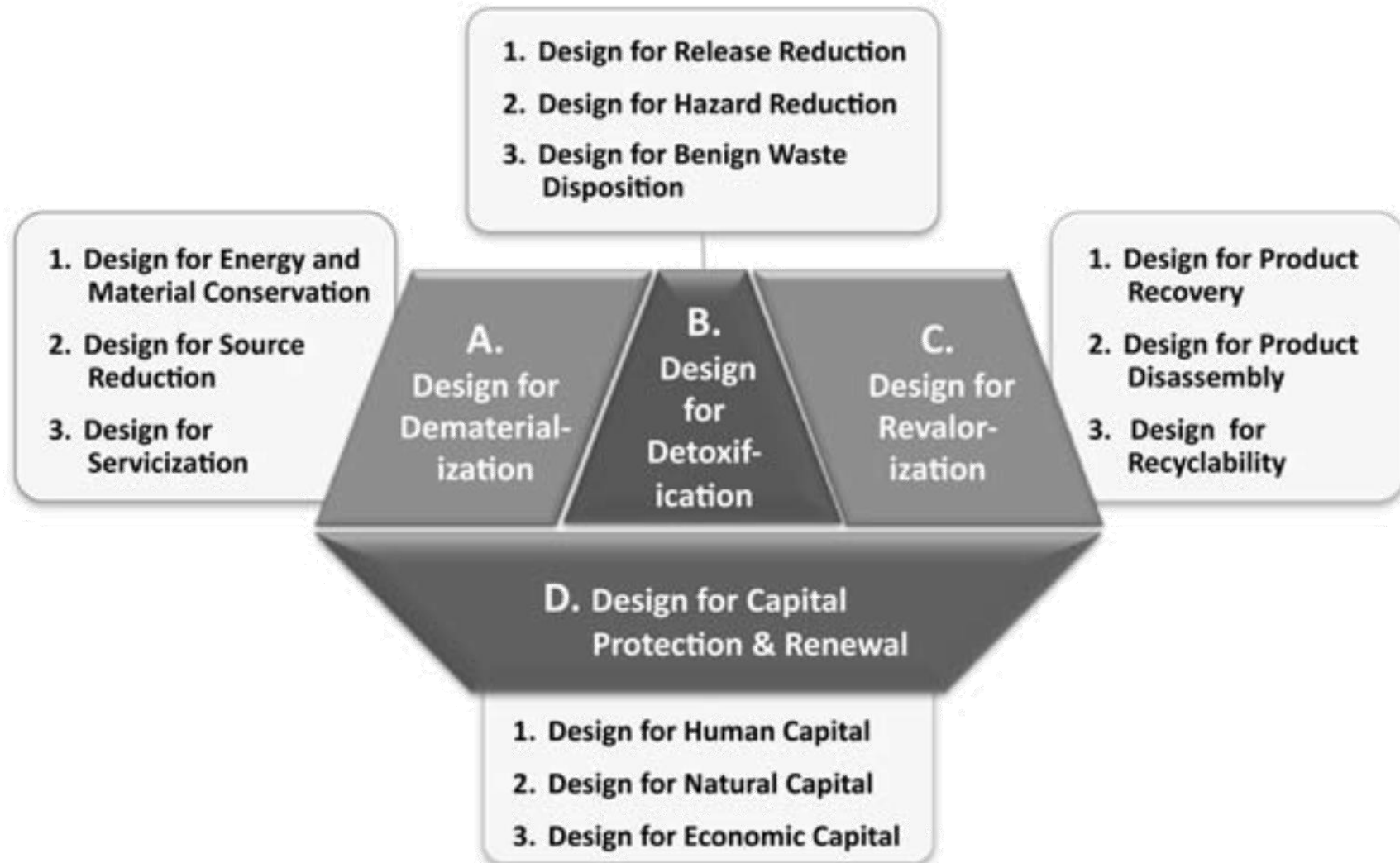
- Performance measurement is a critical to DFE development
- The choice of high-level environmental indicators is extremely important
- With DFE goals, the next step is to decompose the chosen metrics into quantitative parameters that can be tracked throughout production
- A variety of environmental performance measures are used, reflecting regional and industry-specific environmental issues.
- E.g., metrics include resource utilisation measures, atmospheric emission measures, and waste minimisation measures.

4. Design Strategies

Build upon past experiences to assemble a portfolio of design strategies that can be codified, communicated through training, and systematically applied by your design teams.

- i. Design for Dematerialisation
- ii. Design for Detoxification
- iii. Design for Revalorization
- iv. Design for Capital Protection and Renewal

Principal Design Strategies



5. Analysis Methods

- **Screening** methods are used to narrow design choices among a set of alternatives
- **Performance assessment** methods are used to estimate the expected performance of designs for particular indicators
- **Trade-off analysis** methods are used to compare the expected cost and performance of several design alternatives

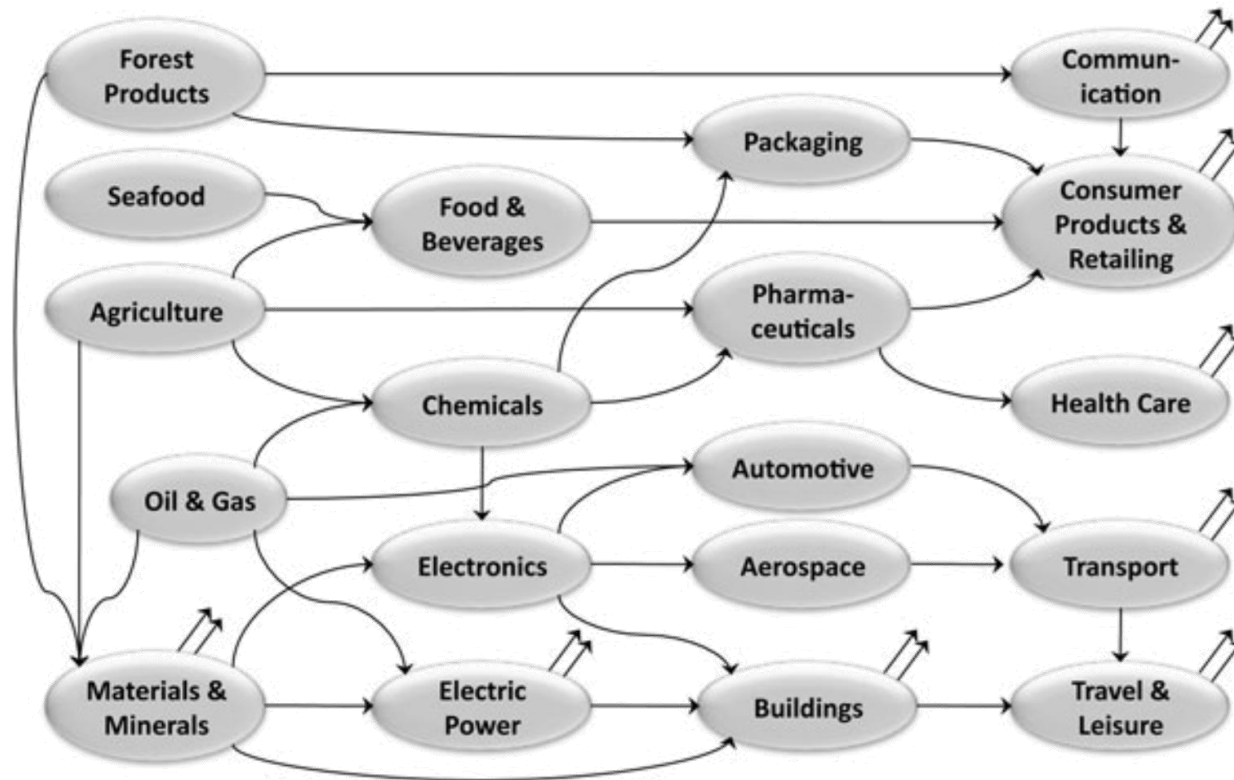
6. Software to Support DFE

Common examples include:

- **Online advisory tools** can help engineers to benefit from accumulated wisdom about design guidelines, useful technologies, or lessons learned from previous successes and failures
- **Computer-based modeling** can estimate DFE consequences because of the complexity of the life-cycle systems associated even with simple products (e.g., carbon footprint calculations)
- **Linkage to the CAE/CAD framework** can encourage regular use of DFE tools by enabling seamless application as part of the normal workflow

7. Inspiration from Nature

Try to emulate the sustainability and resilience of natural systems in designing product and process technologies



Industrial ecosystems convert natural resources into economic goods and services.

Summary

- A successful DFE program starts with top-level buy-in and targets, followed by a plan for continuous improvement
- The seven principles of DFE applications help to define what to implement and how to move forward
- Understanding, analysing and evaluating engineering design across the entire lifecycle (time and space) is critical